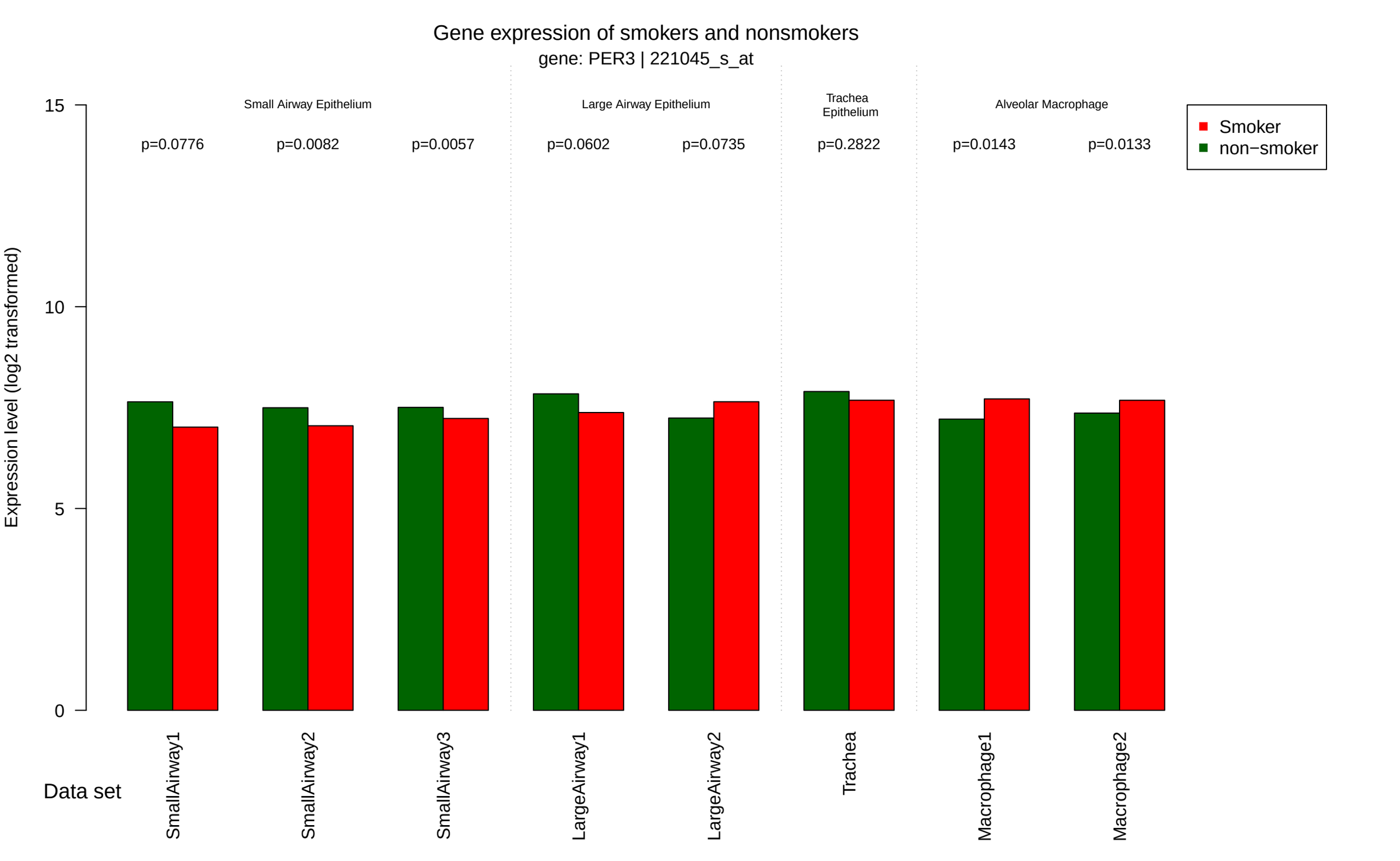
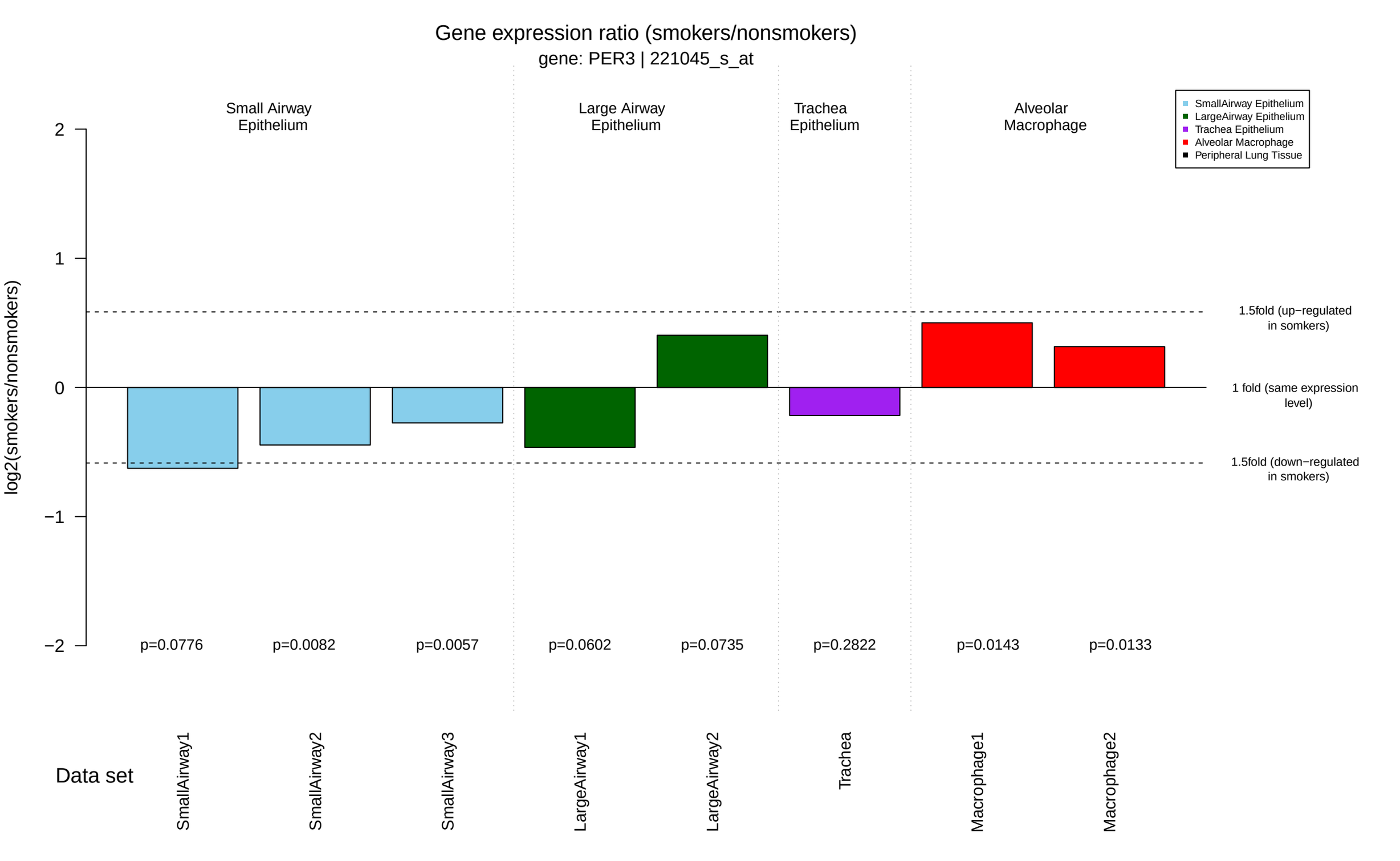
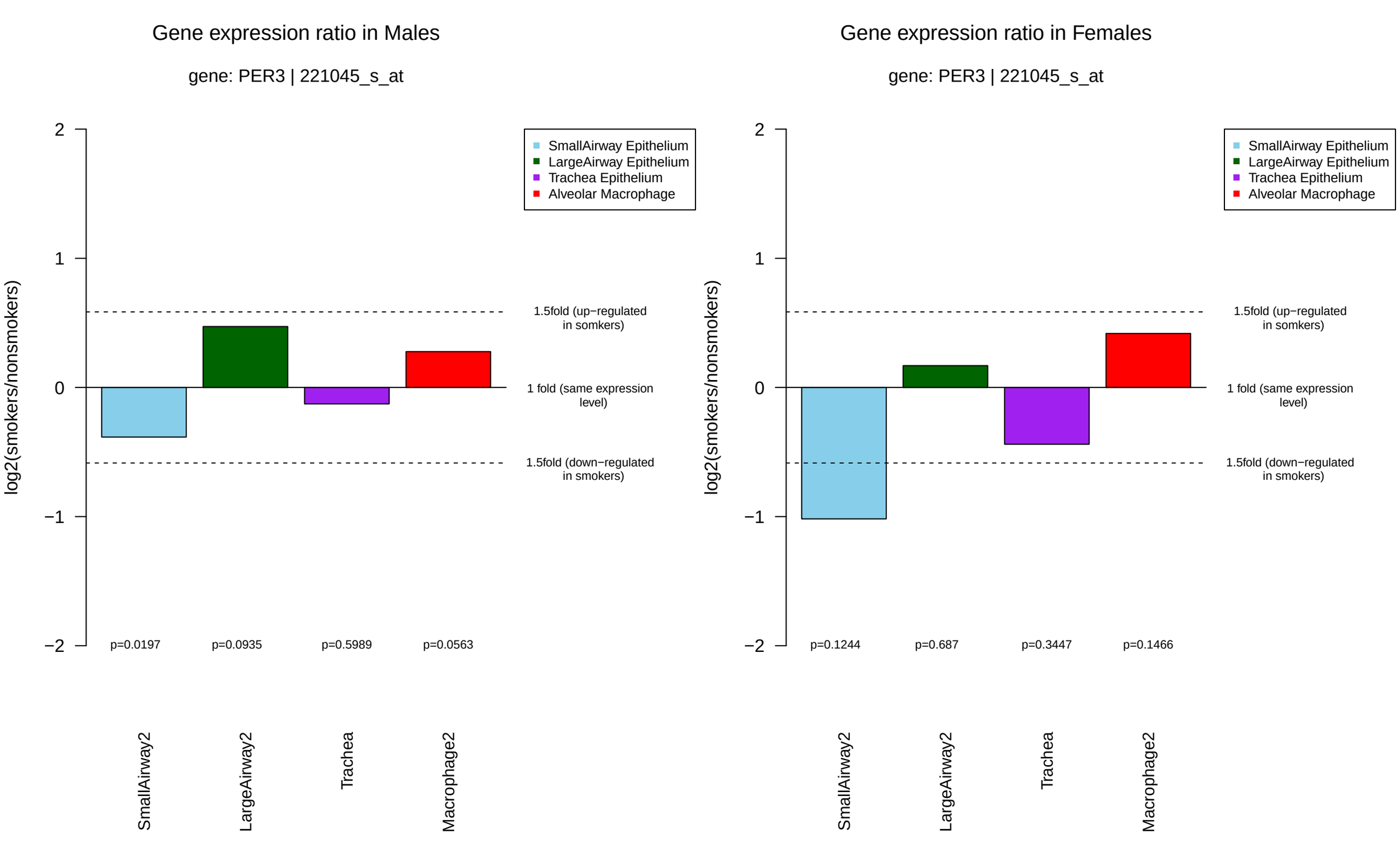
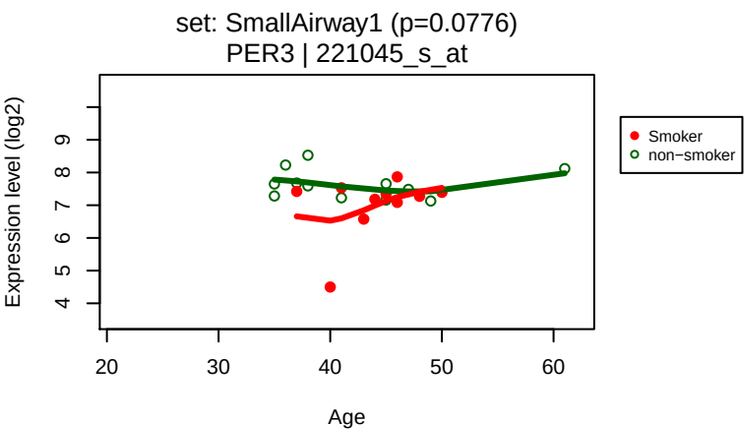
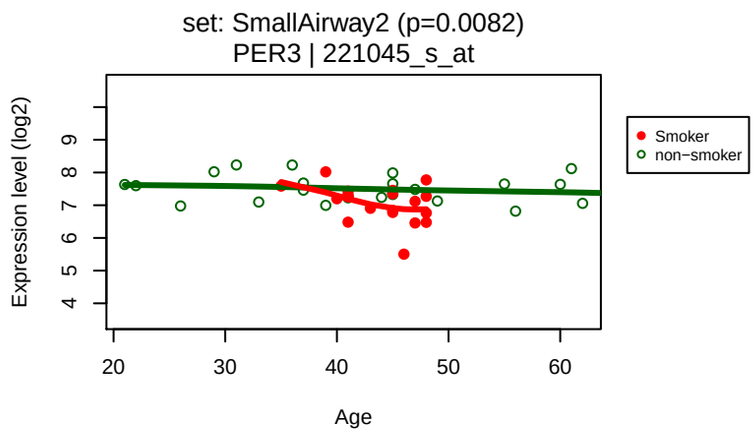
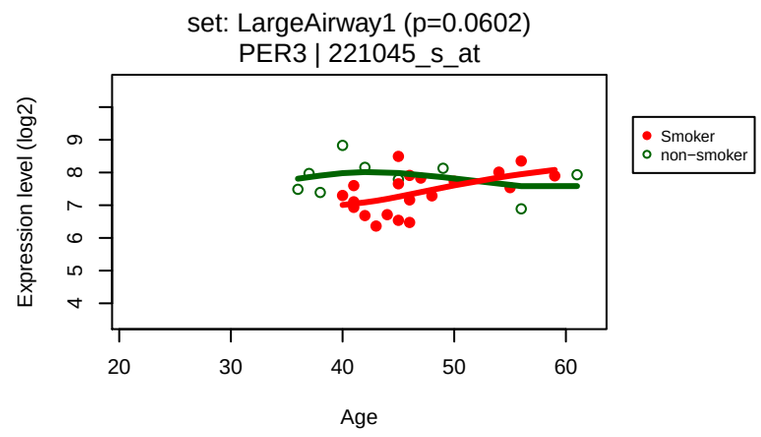

Gene expression of smokers and nonsmokers
gene: PER3 | 221045_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0776
p=0.0082
p=0.0057
p=0.0602
p=0.0735
p=0.2822
p=0.0143
p=0.0133
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

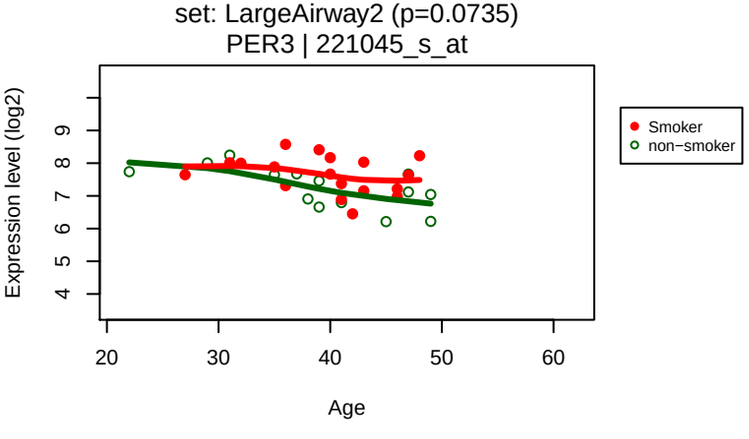

Gene expression ratio (smokers/nonsmokers)
gene: PER3 | 221045_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0776
p=0.0082
p=0.0057
p=0.0602
p=0.0735
p=0.2822
p=0.0143
p=0.0133
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

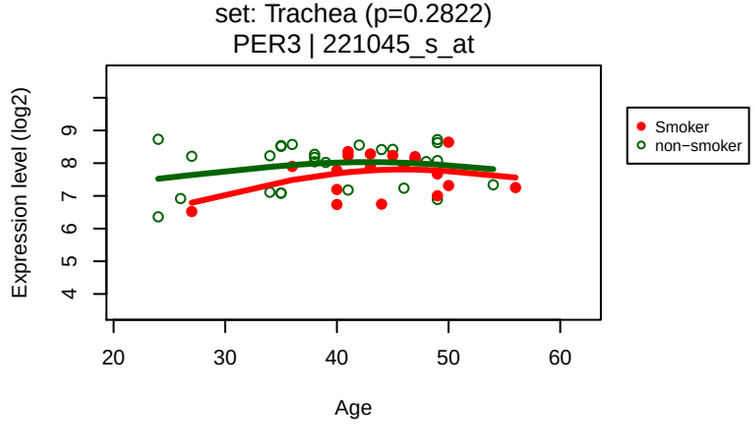

Gene expression ratio in Males
gene: PER3 | 221045_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0197
p=0.0935
p=0.5989
p=0.0563
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PER3 | 221045_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1244
p=0.687
p=0.3447
p=0.1466
SmallAirway2
LargeAirway2
Trachea
Macrophage2

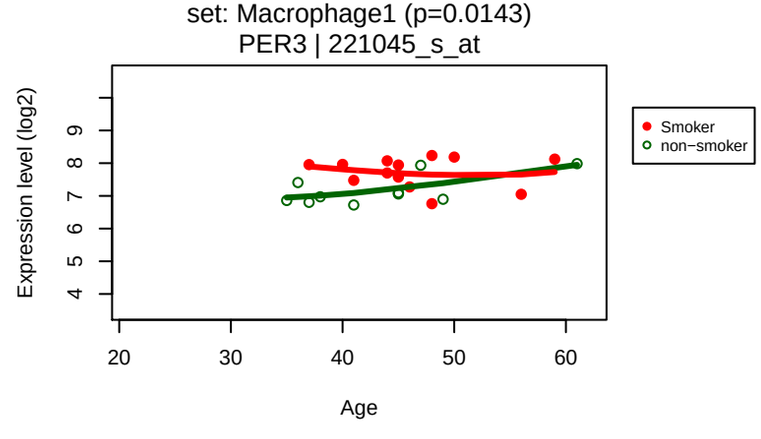

set: SmallAirway1 (p=0.0776)
PER3 | 221045_s_at
Expression level (log2)
Age
Smoker
non−smoker

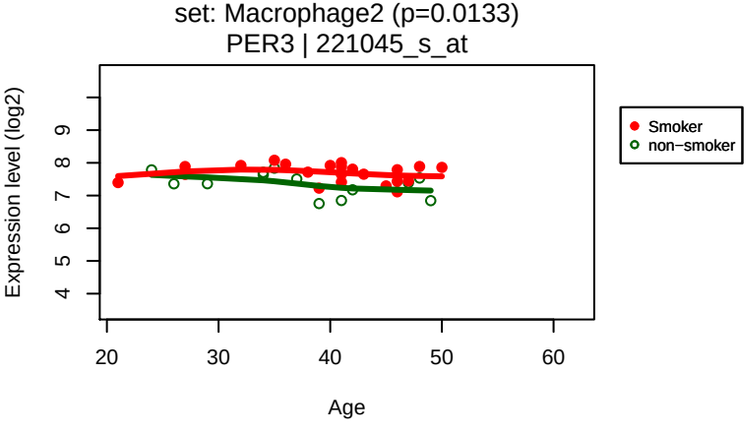

set: SmallAirway2 (p=0.0082)
PER3 | 221045_s_at
Expression level (log2)
Age
Smoker
non−smoker

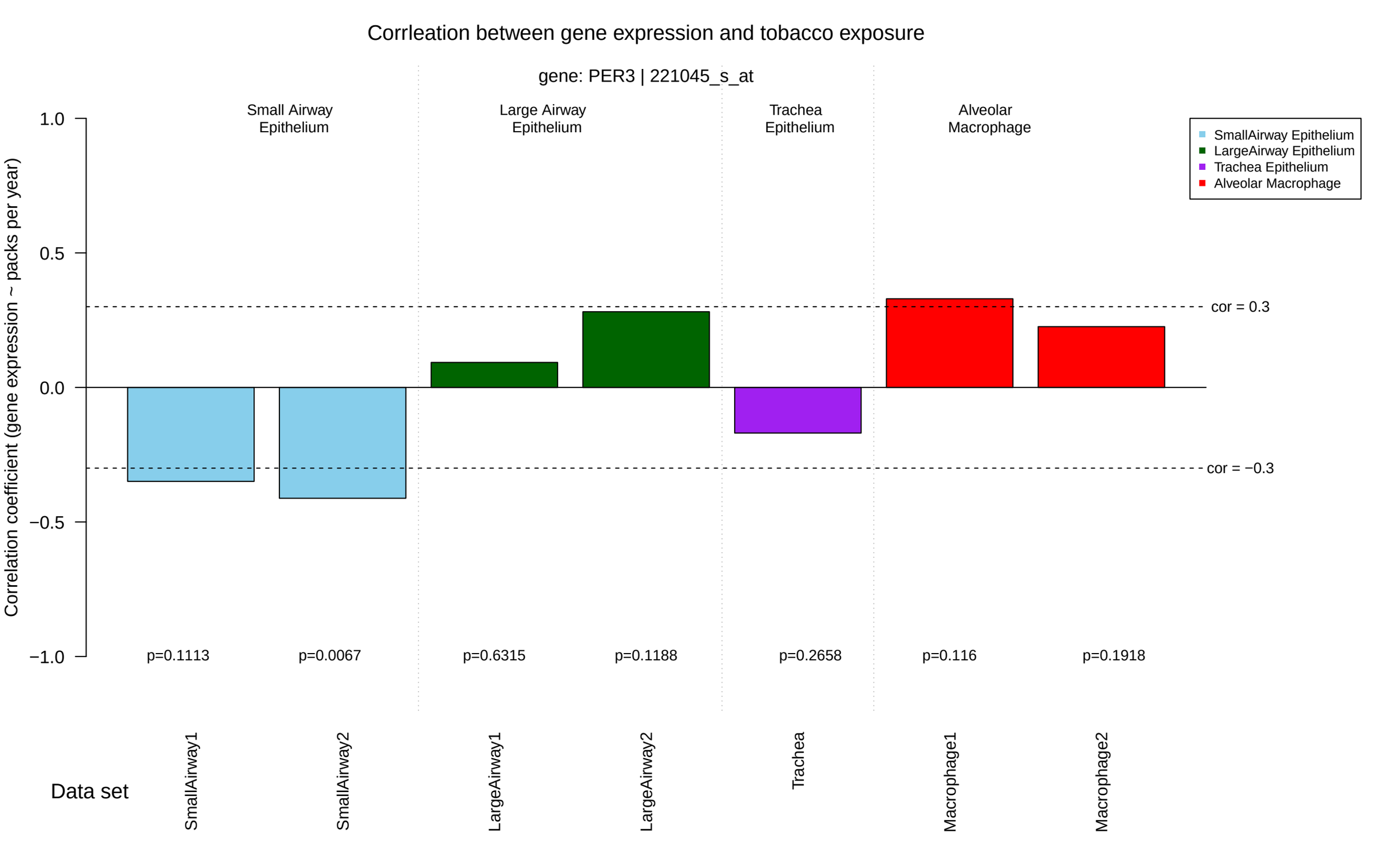

set: LargeAirway1 (p=0.0602)
PER3 | 221045_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0735)
PER3 | 221045_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2822)
PER3 | 221045_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0143)
PER3 | 221045_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0133)
PER3 | 221045_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PER3 | 221045_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1113
p=0.0067
p=0.6315
p=0.1188
p=0.2658
p=0.116
p=0.1918
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2